Satyam Patel

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SUMMARY

Associate AI Engineer with early-career expertise designing scalable AI systems and end-to-end ML pipelines. Delivered a Retrieval-Augmented Generation architecture that reduced LLM hallucinations by 98% and built a vector-search solution handling 25 k documents with sub-1.5 s latency; also improved human pose estimation accuracy by 18% through data refinement and advanced annotation pipelines. Seeking to leverage these results to accelerate AI product development and drive measurable business impact.

EDUCATION

University of Alberta — **Apr 2026**
Specialization in Reinforcement Learning — *Edmonton, AB*

Conestoga College — **Dec 2025**
Post Graduate Program, Big Data Solution Architecture — *Kitchener, ON*

Conestoga College — **Aug 2024**
Post Graduate Program, Artificial Intelligence and Machine Learning — *Waterloo, ON*

Lovely Professional University — **Dec 2020**
Bachelors of Technology, Computer Science — *India*

EXPERIENCE

Wellfound

Client 1: Computer Vision Data Specialist — **Dec 2025 - Present**

- Boosted human pose estimation accuracy by 18% in real-world environments by identifying and resolving over 1,000 complex occlusion edge cases within the training dataset.
- Refined spatial alignment for ground-truth data across 10,000+ images by recalibrating overlapping body keypoints with in-house script tools, which improved input fidelity for the computer-vision pipeline
- Decreased false-positive detection rates by 22% through rigorous data quality control and manual keypoint annotation, directly solving the model's inability to handle overlapping limbs.

Client 2: Retrieval-Augmented Generation (RAG) Chatbot Developer — **Jan 2026 - Present**

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) architecture using LangChain and PyTorch on AWS, applying Test-Driven Development and Git version control, which cut LLM hallucinations by 98% and produced factual, context-aware outputs
- Integrated the Pinecone vector database containing over 25,000 embedded client documents, enabling rapid semantic search and retrieving precise domain knowledge in under 1.5 seconds
- Fine-tuned the base language model on proprietary data and deployed the chatbot directly into the client's production application, which drove a 35% increase in user engagement.

Conestoga College

Graduate Research Assistant — *Waterloo, ON* · **Jan 2024 - Aug 2024**

- Engineered "The Plant Guard" system and conducted academic research to address precision-agriculture challenges by designing end-to-end computer-vision pipelines with TensorFlow and OpenCV, enabling early detection of plant diseases and reducing manual scouting time
- Processed and augmented a dataset of tens of thousands of images using Python and Pandas, then delivered the cleaned data to leadership so they could make informed decisions
- Architected and trained multiple neural networks using TensorFlow and Keras, establishing a baseline with a custom Vanilla CNN and improving feature extraction by implementing Transfer Learning with a fine-tuned VGG16 model.
- Achieved a 90 percent + classification accuracy across multiple plant species and pathology categories, delivering a scalable, modular machine learning pipeline optimized for future deployment on mobile and edge computing devices.

PROJECTS

Autonomous LLM Reasoning Agent (RLVR / GRPO) PyTorch, GRPO, LoRA, Unsloth, Hugging Face

- Drove advancements in LLM technologies by fine-tuning new foundational models for autonomous chain-of-thought reasoning.
- Oversaw the deployment, scaling, and inference of the optimized ML solutions on constrained hardware using LoRA and 4-bit quantization.
- Optimized training efficiency by utilizing 4-bit quantization, LoRA, and Unsloth, successfully executing a full RL pipeline on constrained GPU hardware
- Built deterministic experimentation frameworks to evaluate model outputs, demonstrating adaptability to learn and evolve as newer AI architectures emerge.

Multi-Agent Text-to-SQL Data Analyst LangGraph, Llama-3.3(Groq), MongoDB, Docker

- Migrated a multi-agent Generative AI application to a self-hosted **n8n** server, automating complex **LangGraph** LLM workflows to translate natural language into SQL, which reduced manual query processing time by 85%.
- Re-architected the backend infrastructure by transitioning to a highly scalable **MongoDB** database, optimizing the system to handle enterprise datasets and supporting 50,000+ concurrent real-time requests with sub-second latency.

- Productionized real-time AI microservices using **FastAPI** and **Docker**, engineering automated CI/CD pipelines that accelerated model deployment cycles by 40% while ensuring robust data governance and security.
- Implemented an automated CI/CD-ready workflow and engineered startup scripts to ensure data quality, governance, and security across relational databases.

Autonomous Cloud Resource Optimizer Stable-Baselines3 (PPO), Gymnasium, Streamlit, TensorBoard, Plotly
- Engineered a custom Reinforcement Learning environment using Python and Gymnasium to simulate network traffic, and trained a PPO agent with Stable-Baselines3 to autonomously scale cloud servers up or down based on demand.
- Designed a specialized reward function that balanced system performance with power efficiency, successfully reducing simulated cloud infrastructure costs by 30% while maintaining reliable server uptime.
- Packaged the trained model into a Dockerized FastAPI microservice with RESTful APIs, creating a ready-to-deploy CI/CD solution that can easily integrate with platforms like Microsoft Azure and AWS for real-time resource management.

Retail CCTV Security & Voice Assistant AI YOLOv8x, YuNet/SFace, WebSocket, OpenAI TTS, OpenCV
- Engineered an end-to-end CCTV processing pipeline using YOLOv8x and YuNet/SFace ONNX models, enabling automated facial embedding extraction and centroid-based multi-person tracking across retail footage.
- Built a FastAPI backend with asynchronous job threading and WebSocket alert channels, delivering real-time watchlist match notifications and live annotated stream previews to an operator dashboard without polling.
- Developed an interactive store phone assistant powered by OpenAI TTS and a local LLM, utilizing multi-turn context memory and mood-aware prosody to handle real-time inventory and product lookups
- Designed a modular REST and WebSocket API surface with session management, codec-aware video output, and automatic ONNX model bootstrapping on the first run to minimize deployment friction.

AI Automation Pipeline Suite n8n, Google Cloud Vertex AI, Veo, Gemini, Claude API, FFmpeg
- Built an end-to-end AI video production pipeline in n8n that converts a single text prompt into a 12-scene storyboard, AI-generated video clips, captions, narration assets, and render-ready outputs using Gemini 2.5 Pro and Veo.
- Architected a resilient asynchronous orchestration layer for long-running generation jobs with polling, retry logic, cloud status tracking, GCS URI-to-HTTPS conversion, and automated asset handling, keeping per-clip delivery under 3 minutes.
- Automated post-production and publishing by generating SRT caption files, FFmpeg concat manifests, structured output paths, quality-filtered downloads, and YouTube uploads with metadata, privacy settings, and video tracking handled inside n8n.
- Developed a multi-brand AI product feed optimization system using LLM APIs to process full product catalogs and generate Google Merchant Center titles, Google Ads copy, Meta ad assets, and campaign-ready marketing files from a centralized brand configuration.

CERTIFICATIONS

- Microsoft AI & ML Engineering: Microsoft Azure for AI and Machine Learning
- Microsoft Building Intelligent Troubleshooting Agents: Microsoft Azure AI: Workloads and Machine Learning on Azure
- TCPS 2: CORE
- Microsoft Professional Program in Artificial Intelligence
- COC Workship
- Amazon Alexa Developer

TECHNICAL SKILLS

- **Generative AI & LLMs**: LangChain, LangGraph, Vector Databases, RAG Systems, OpenAI APIs, Open-Source LLMs (DeepSeek, Llama-3.3), Prompt Engineering, LoRA, Hugging Face, Foundational Models, Agentic AI, Google Gemini, Vertex AI, Veo, Text-to-Speech (TTS)
- **Machine Learning & Deep Learning**: PyTorch, TensorFlow, Keras, Scikit-Learn, NLP, Feature Engineering, Statistical Modeling, Anomaly Detection, Data Mining, Model Drift Monitoring, Deep Reinforcement Learning
- **Reinforcement Learning**: PPO, GRPO, Stable-Baselines3, Custom MDP Design, Power Efficiency Modeling, Gymnasium, Custom SARSA
- **Data Engineering & Languages**: Python (NumPy, Pandas, Scikit-learn), SQL (PostgreSQL, SQLite), C/C++, Apache Spark, Hadoop, Data Pipelines, Data Structures & Algorithms, JavaScript
- **MLOps & Cloud**: Docker, FastAPI (Microservices), RESTful APIs, CI/CD Pipelines, Microsoft Azure, AWS, Streamlit, TensorBoard, Git, Linux, DevOps, Test-Driven Development (TDD), Google Cloud Platform (GCP), Google Cloud Storage (GCS)
- **XAI & Interpretability**: Visualization Techniques, Reward/Feature Attribution, SHAP, LIME
- **Databases & Big Data**: NoSQL Ecosystems, PostgreSQL, SQLite, Apache Spark, Hadoop
- **AI Orchestration & Automation**: n8n, Multi-Agent Workflows, Agentic AI, CI/CD Automation, Apache Kafka / RabbitMQ, Semantic Routing / Tool Calling, ETL / ELT Automation, Webhooks & API Integration, YouTube Data API v3, RapidAPI, OAuth 2.0 Integration